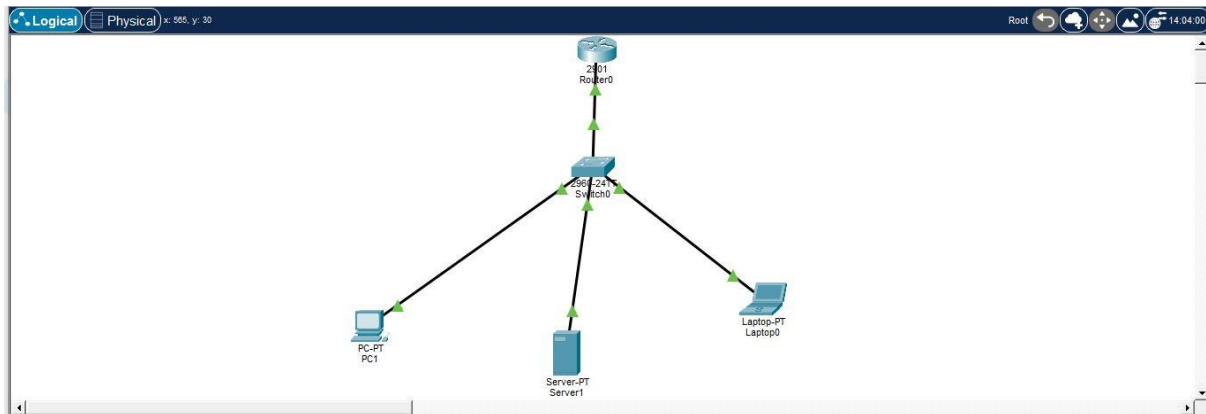
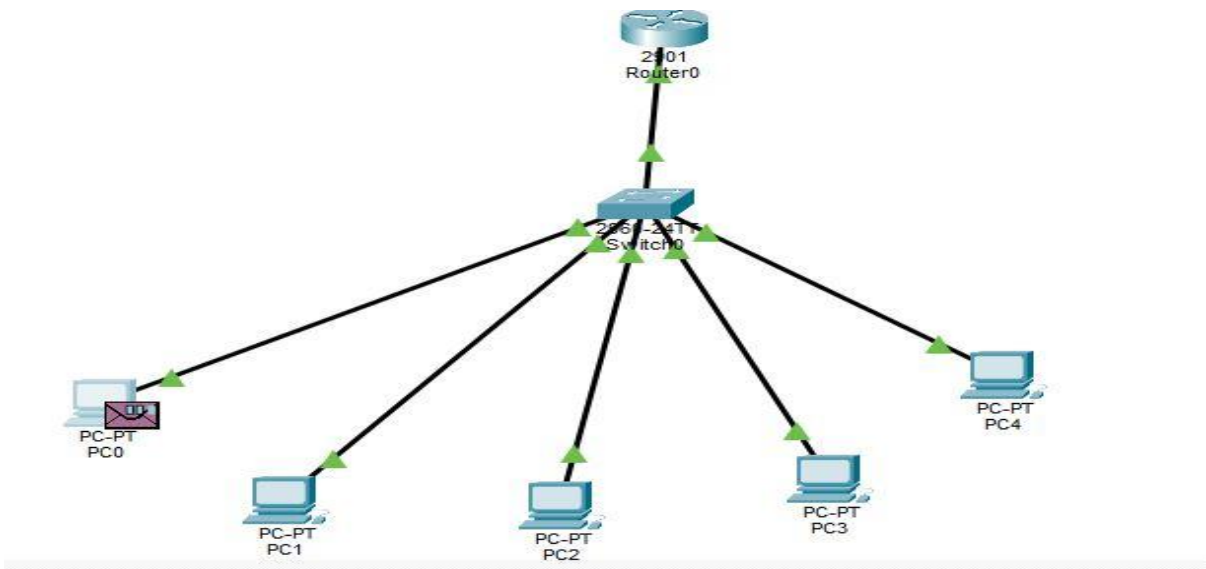


1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

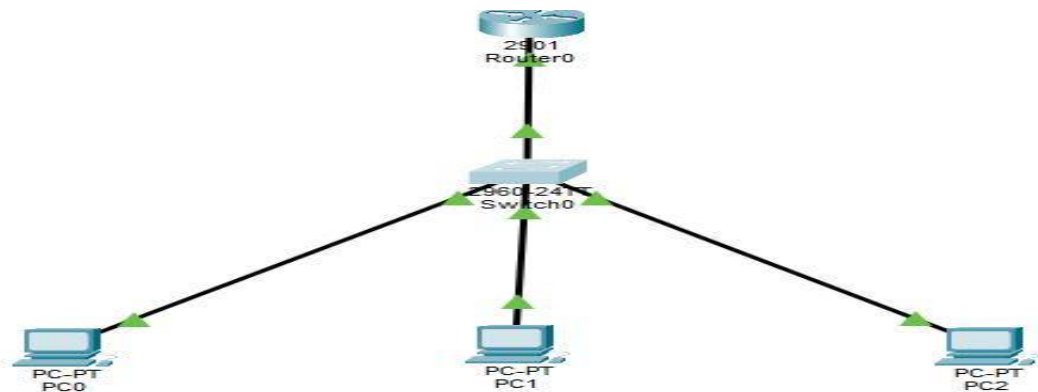


2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.



- Here, in this network. I have created two VLANs: VLAN 10 and VLAN 20. In the VLAN 10 i have 3 devices and in VLAN 20 I have 2 devices. Then, I tried to send request by using ping from VLAN 10's one device to VLAN 20's one device. However, it's not happening because there two VLANs in the same network.

3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.



- Here, I have created two VLANS, VLAN 10 and VLAN 20. In VLAN 10, I have two devices and in VLAN 20 I have one device. Then, I tried to ping from first device of VLAN 10 to another device of VLAN10 and it was communicating successfully.

Switch0

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
10	SALES						active			
20	HR						active			
1002	fddi-default						active			
1003	token-ring-default						active			
1004	fddinet-default						active			
1005	trnet-default						active			
1	enet	100001	1500	-	-	-	-	-	0	0

- Then, I moved one device from VLAN10 to VLAN20 and configure it and it configure successfully.

4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

- Here, I am using short script for VLAN 30 in Cisco Packet Tracer.
- En
- Conf t
- Vlan 30
- Name VLAN30
- Exit
- Interface range f0/1-10
- Switchport mode access
- Switchport access Vlan 30
- No shutdown
- Exit
- End
- Write memory